Context-Length Lower Bounds for Reference Resolution in Transformer Language Models

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Abstract

Long-context language models are increasingly deployed on tasks requiring coherent discourse reasoning, yet evaluation methodology remains dominated by retrieval and span extraction metrics. These metrics fail to isolate the structured linguistic computations required for discourse reference resolution: the process of linking referring expressions to their antecedents under entity competition, salience dynamics, and multi-level anaphoric chaining. We argue that this failure produces systematically misleading conclusions about model capability.

We introduce a formal framework modeling discourse reference as a depth-sensitive, interference-sensitive state-tracking problem. We define *anaphoric depth* k as the number of chained resolution steps required to identify a terminal referent, and *distractor interference* η as the number of competing candidates introduced per chain level. Modeling transformer attention as a mechanism with bounded effective support, we derive that resolving reference chains of depth k with η distractors per level requires effective attention support of size $\Omega(k \cdot \eta)$, a bound that cannot be circumvented by increasing nominal context length without architectural changes.

We construct RefChain, a controlled benchmark of 4,800 discourse passages spanning four reference types (pronouns, definite descriptions, bridging anaphora, discourse deixis), six depth levels ($k \in \{1, \ldots, 6\}$), and four interference levels ($\eta \in \{1, 2, 3, 4\}$), with discourse graph structure controlled independently of lexical content. Evaluating eleven language models and four baselines, we observe structured degradation whose breakpoints align with theoretical predictions ($R^2 = 0.87$ on aggregate). Critically, errors correlate with discourse depth and interference structure rather than with token distance or lexical rarity, directly supporting the architectural explanation over positional or statistical explanations.

This work provides the first linguistically grounded theoretical account of long-context degradation, reframes benchmark design as a discourse-theoretic rather than retrieval-theoretic problem, and yields testable predictions for future architectural development.

Keywords: discourse reference resolution, coreference, long-context language models, formal lower bounds, transformer expressivity, anaphoric depth, benchmark construction

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1 Introduction

1.1 The Problem with Long-Context Evaluation

The extension of transformer context windows has been one of the defining engineering developments of the current language model generation. Models supporting 32K, 128K, and even million-token contexts have been released in rapid succession (**???**). The accompanying evaluation literature has responded with increasingly long benchmarks: needle-in-a-haystack retrieval tests, long-document QA, and cross-document summarization (**???**).

These evaluations share a fundamental structural assumption: that processing long context is primarily a retrieval problem. A model succeeds by locating and extracting a relevant passage; it fails by overlooking or confusing it with irrelevant content. The "lost in the middle" phenomenon (**?**), which documents performance degradation for facts placed in the center of long contexts, has been widely cited as the defining long-context failure mode.

We argue this framing is incomplete. Discourse comprehension is not retrieval. Natural language understanding requires maintaining structured representations of entities, events, and relations across text, tracking how those representations update as new information is introduced, and resolving linguistic expressions to their appropriate discourse objects under ambiguity. These operations are governed by discourse structure rather than position, and they fail for reasons that no retrieval-oriented evaluation can diagnose.

1.2 Discourse Reference as the Test Case

Reference resolution is the canonical instance of structured discourse computation. Resolving a pronoun requires identifying its antecedent among competing candidates while accounting for recency, salience, syntactic constraints, and semantic plausibility (**?**). Resolving a definite description such as *the director* requires constructing an entity representation from prior discourse and linking it to the current expression under world knowledge. Bridging anaphora (**?**) require inferring implicit discourse relations. Discourse deixis (**?**) requires abstracting over propositional content.

None of these operations decompose into retrieval. They require maintaining and updating entity state across intervening context, tracking which entities remain salient, and resolving structured ambiguity. When reference chains are nested—when the identity of a referent depends on resolving another referent, which depends on resolving a third—the computation becomes hierarchically complex in a way that no positional analysis captures.

Consider the following constructed example:

Example 1.1 (Depth-3 reference chain with interference)**.** *"Dr. Hartley introduced Dr. Vance to the committee. The physicist from Berlin had supervised her doctoral work. After the conference, she sent him a copy of the manuscript that he had first described to her in Geneva. The claim that it contained had originally been developed by the person she had thanked in her acknowledgments."*

Resolving the final noun phrase *the person she had thanked in her acknowledgments* requires (1) identifying *she* as Dr. Vance, (2) identifying Dr. Vance's antecedent in the first sentence, (3) reasoning about who she thanked based on discourse structure. The correct resolution cannot be retrieved; it must be computed through a chain of prior resolutions, each of which is subject to interference from competing mentions.

1.3 The Formal Gap

Despite extensive observation of long-context failure, there is no formal account linking discourse structure to architectural limitations. "Lost in the middle" is a symptom, not an explanation. It describes where models fail but not why, and it provides no prediction about what structural properties of input make failure more likely.

The absence of a formal account has two practical consequences. First, benchmark design remains atheoretical: evaluators measure aggregate accuracy on long documents without controlling for the discourse properties that predict difficulty. Second, architectural design lacks principled guidance: there is no framework connecting attention mechanism properties to the discourse computations they can or cannot support.

1.4 Contributions

This paper makes the following contributions:

(i) **Formal framework.** We model discourse reference as a constrained state-tracking process with explicit depth and interference parameters and define effective attention support as the mechanism connecting transformer architecture to resolution capacity (Section 3).

(ii) **Lower bound theorems.** We prove that resolving reference chains of depth k with η distractors per level requires $\Omega(k \cdot \eta)$ effective attention support, and that under positional decay (as in RoPE and ALiBi), interference-driven degradation follows a predictable scaling law (Section 3).

(iii) **Benchmark construction.** We introduce RefChain, a 4,800-instance controlled discourse benchmark spanning four reference types, six depth levels, and four interference levels, with discourse graph structure varying independently of surface lexical properties (Section 4).

(iv) **Multi-model empirical evaluation.** We evaluate eleven language models and four baselines on RefChain, producing per-model accuracy profiles as a function of depth and interference (Section 5).

(v) **Theoretical–empirical alignment.** We demonstrate that empirical breakpoints align with theoretical predictions ($R^2 = 0.87$), and that error distributions correlate with discourse structure rather than token distance (Section 6).

(vi) **Architectural implications.** We show that context window size alone does not predict discourse competence, and that chain-of-thought prompting shifts but does not eliminate the depth bound (Section 6).

1.5 Paper Organization

Section 2 reviews long-context modeling, discourse theory, and transformer expressivity. Section 3 develops the formal framework and lower bound proofs. Section 4 describes benchmark construction. Section 5 presents experimental setup and results. Section 6 provides analysis and alignment with theory. Section 7 discusses implications and limitations. Section 8 concludes.

2 Background and Related Work

2.1 Long-Context Language Models

Transformer language models (**?**) process sequences through self-attention, which in its standard form has quadratic complexity in sequence length. Practical extensions to long contexts have taken two directions: positional encoding modifications and attention sparsification.

Positional encodings. Absolute positional encodings (**?**) degrade under length extrapolation. Relative encodings including Rotary Position Embedding (RoPE) (**?**) and ALiBi (**?**) embed positional information as attention biases and generalize better to unseen lengths. NTK-aware interpolation (**?**) and YaRN (**?**) extend these further. Critically, all of these encodings introduce positional decay: attention weights decrease as a function of distance, either through frequency compression or explicit penalty. This decay is the mechanism our lower bound exploits.

Sparse attention. Longformer (**?**) and BigBird (**?**) reduce complexity through sliding window and random attention patterns. These architectures have bounded effective receptive fields for individual positions, making the support constraints in our framework directly applicable.

Evaluation. Long-context evaluation has been dominated by SCROLLS (**?**), ZeroSCROLLS (**?**), the Long Range Arena (**?**), QuALITY (**?**), NarrativeQA (**?**), and needle-in-a-haystack probing (**?**). RULER (**?**) introduces multi-hop retrieval but does not formalize discourse graph structure. **None of these benchmarks control anaphoric depth or distractor interference as independent variables.** Our work fills this gap.

2.2 Discourse Reference and Centering Theory

Centering. The Centering Theory of **?** formalizes entity salience in discourse. It defines the *forward-looking centers* of an utterance as the entities evoked and the *backward-looking center* as the most salient entity. Transitions between utterances are typed by how centers shift, predicting coherence and referential form. We operationalize salience as a parameter in our discourse state model.

Reference types. Coreference (**?**) covers pronouns and definite descriptions. Bridging anaphora require inferring an implicit associative relation (e.g., *the engine* from *a car*). Discourse deixis (**?**)

refers to propositions or events rather than entities. Each type places different demands on entity state maintenance. We include all four in RefChain.

Entity state tracking. Neural models for entity state tracking include **?** and EntityNLI (**?**). Recent work on coreference includes SpanBERT-based systems (**?**), end-to-end neural coref (**?**), and LLM-based approaches (**?**). These systems measure resolution accuracy on naturally occurring text without controlling structural complexity. Our work complements this by providing a framework for predicting when and why systems fail.

2.3 Formal Expressivity of Transformers

? proves that standard transformers cannot represent parity and similar formal language tasks with high probability. **?** characterize the formal languages transformers can recognize in terms of circuit depth. **?** show that saturated attention transformers correspond to AC^0 circuits. **?** and **?** extend this to sequence-to-sequence tasks.

These results establish that transformers have bounded formal expressivity. However, they target formal language tasks (parity, majority, modular arithmetic) rather than linguistic phenomena. Our work is the first to target discourse reference resolution as a formal computation and derive lower bounds from it. The key difference is that our hard instances are defined by linguistically meaningful parameters—anaphoric depth and distractor interference—rather than artificial combinatorial properties.

2.4 Coreference Resolution Systems

Neural coreference resolution has progressed from mention-pair models (**?**) through entity-ranking (**?**) to span-based end-to-end systems (**??**). These systems are typically evaluated on OntoNotes (**?**) and PreCo (**?**). LLM-based coreference (**??**) shows strong performance on naturalistic text.

The key gap relative to our work: existing evaluation measures aggregate performance on naturally occurring text where depth and interference are uncontrolled. Our benchmark separates these variables, enabling diagnosis of the structural source of failure.

2.5 Positioning This Work

Our work sits at the intersection of formal expressivity analysis and discourse linguistics. It differs from long-context benchmarks in that it controls structural complexity theoretically rather than empirically. It differs from transformer expressivity work in that its hard instances are linguistically grounded. It differs from coreference resolution work in that it provides predictive theory rather than engineering improvements.

3 Formal Framework

3.1 Discourse State Model

We model discourse as a sequence of utterances producing an evolving state. Let Σ^* denote the set of all utterance sequences.

Definition 3.1 (Discourse State). *A* discourse state *at time t is a triple:*

$$S_t = (\mathcal{M}_t,\ \sigma_t,\ \rho_t)$$

where:

- $\mathcal{M}_t = \{e_1, \dots, e_n\}$ *is the set of* discourse referents *(entities introduced up to t),*
- $\sigma_t : \mathcal{M}_t \to [0, 1]$ *is a* salience function *assigning prominence scores,*
- $\rho_t : \mathcal{M}_t \times \mathcal{M}_t \to \mathcal{R}$ *encodes discourse relations between entities.*

The state evolves as: $S_{t+1} = \delta(S_t, u_{t+1})$, where u_{t+1} is the next utterance and δ is the state transition function.

Definition 3.2 (Anaphoric Query). *An* anaphoric query *is a tuple $q = (\alpha, \mathcal{C}_q)$ where α is a referring expression and $\mathcal{C}_q \subseteq \mathcal{M}_t$ is the set of candidate antecedents. The* resolution function *$R : \mathcal{C}_q \times S_t \to \mathcal{M}_t$ maps the query to its correct antecedent under the current discourse state.*

3.2 Reference Chains and Depth

Definition 3.3 (Reference Chain of Depth k). *A* reference chain *of depth k is a sequence of utterances $(u_0, u_1, \dots, u_k)$ together with anaphoric queries $(q_1, \dots, q_k)$ such that:*

(1) u_0 introduces a set of entities $\mathcal{E}_0 \subseteq \mathcal{M}_0$,
(2) each q_i has a candidate set $\mathcal{C}_{q_i}$ whose correct resolution depends on correctly resolving q_{i-1},
(3) the terminal query q_k cannot be answered without first resolving all of $q_1, \dots, q_{k-1}$.

The anaphoric depth *is k.*

Definition 3.4 (Distractor Interference). *At each chain level i, let η_i be the number of* distractor entities*: entities in $\mathcal{C}_{q_i}$ that are not the correct antecedent but share salient properties with it. We define the* interference parameter *$\eta = \max_i \eta_i$, and the* total candidate pressure *at level i as $|\mathcal{C}_{q_i}| = \eta_i + 1$.*

The key observation is that distinguishing the correct antecedent from η distractors requires accessing the specific distinguishing evidence that was introduced at the corresponding chain level, which may be arbitrarily distant from the query position.

3.3 Transformer Attention Model

Definition 3.5 (Effective Attention Support). *For a transformer $\mathcal{T}$ with attention weight function $\alpha_{t,j}$ at query position t attending to key position j, and a threshold $\epsilon > 0$, the* effective attention support *is:*

$$\mathcal{A}_\epsilon(t) = \{ j < t \mid \alpha_{t,j} > \epsilon \}$$

$|\mathcal{A}_\epsilon(t)|$ is the number of positions usable by position t for computing its output.

Definition 3.6 (Positional Decay). *A positional encoding scheme exhibits* positional decay *if, for any layer ℓ and head h, the attention weight satisfies:*

$$\mathbb{E}[\alpha_{t,j}^{(\ell,h)}] \leq f(t-j)$$

for some function f with $f(d) \to 0$ as $d \to \infty$. Both RoPE and ALiBi satisfy this with $f(d) = O(d^{-\beta})$ for encoding-class-dependent $\beta > 0$.

3.4 Indistinguishability Lemma

Lemma 3.7 (Indistinguishability under Support Constraint). *Let e^* be the correct antecedent for query q and let $e^\dagger$ be a distractor sharing the same surface form as e^*. If the distinguishing evidence v for e^* over $e^\dagger$ satisfies $v \notin \mathcal{A}_\epsilon(t_q)$, then for any transformer $\mathcal{T}$, the representations h_{e^*} and $h_{e^\dagger}$ at the query position are indistinguishable, and the model resolves the query correctly with probability at most $\frac{1}{|\mathcal{C}_q|}$.*

Proof. The output of position t_q in layer ℓ is:

$$h_{t_q}^{(\ell)} = \sum_{j \in \mathcal{A}_\epsilon(t_q)} \alpha_{t_q,j}^{(\ell)} V_j^{(\ell)} + \text{residual}$$

Since $v \notin \mathcal{A}_\epsilon(t_q)$, the distinguishing signal V_v does not contribute to the output. The representations of e^* and $e^\dagger$ that are accessible differ only through information shared by both. The model therefore cannot prefer e^* over $e^\dagger$ based on content, reducing the task to an uninformed choice among $|\mathcal{C}_q|$ candidates. □

3.5 Main Lower Bound

Theorem 3.8 (Depth–Interference Lower Bound). *For a reference chain of depth k with interference parameter η, any causal transformer $\mathcal{T}$ requires:*

$$|\mathcal{A}_\epsilon(t_k)| \;\geq\; \Omega(k \cdot \eta)$$

to resolve the terminal query q_k correctly with probability $\geq 1 - \delta$ for any fixed $\delta < 1 - \frac{1}{\eta+1}$.

Proof. We proceed by induction on depth k.

Base case ($k = 1$). The query q_1 has $\eta + 1$ candidates. By Lemma 3.7, each of the η distractors requires distinct distinguishing evidence at distinct positions. To distinguish e^* from all η distractors with success probability $\geq 1 - \delta$, the support must contain at least η distinguishing evidence positions. Thus $|\mathcal{A}_\epsilon(t_1)| \geq \eta = \Omega(\eta)$.

Inductive step. Assume the claim holds for chains of depth $k-1$. A depth-k chain is a depth-$(k-1)$ chain followed by an additional resolution step q_k. Correctly resolving q_k requires (a) correctly resolving the preceding chain to obtain the state S_{k-1} from which q_k's correct antecedent can be identified, and (b) accessing the distinguishing evidence for q_k among its η distractors.

By the inductive hypothesis, step (a) requires support of size $\Omega((k-1)\cdot\eta)$, and those positions must remain within $\mathcal{A}_\epsilon(t_k)$ because the model must "remember" the intermediate resolutions. Step (b) adds η new positions. The total is $\Omega((k-1)\cdot\eta) + \Omega(\eta) = \Omega(k\cdot\eta)$. □

Corollary 3.9 (Context Length Does Not Suffice)**.** *Increasing the nominal context window of $\mathcal{T}$ without increasing $|\mathcal{A}_\epsilon(t_k)|$ does not improve resolution accuracy on depth-k chains. Effective support, not context length, is the binding constraint.*

Theorem 3.10 (Interference Scaling Law under Positional Decay)**.** *Under positional decay (Definition 3.6) with rate β, for a fixed total document length L, the expected accuracy on a depth-k, interference-η chain satisfies:*

$$\mathbb{E}[\mathrm{Acc}(k,\eta)] \;\leq\; \exp\Big(-\alpha\cdot k\cdot\eta\cdot L^{-\beta/d}\Big)$$

for constants $\alpha > 0$ and d depending on the encoding class.

Proof Sketch. Under positional decay, the probability that distinguishing evidence at distance D is in $\mathcal{A}_\epsilon(t_k)$ is bounded by $f(D)\cdot\epsilon^{-1}$. For a chain of depth k with η distractors per level, where evidence is placed adversarially at maximum depth, the joint probability of accessing all $k\eta$ required positions decays multiplicatively. Taking the product over independent access events and applying the decay bound yields the exponential form. The full derivation is provided in Section A. □

Proposition 3.11 (Chain-of-Thought Cannot Eliminate the Bound)**.** *For any chain-of-thought prompting strategy π, if $\mathcal{T}$ has effective support $|\mathcal{A}_\epsilon(t)| < c\cdot k\cdot\eta$ for constant $c < 1$, then there exist reference chains of depth k with interference η on which $\pi(\mathcal{T})$ fails with probability at least $\delta_0 > 0$ regardless of the prompt.*

Proof. Chain-of-thought prompting generates intermediate tokens that can serve as scratchpad memory, effectively extending the input. However, the generation of each intermediate token is subject to the same attention support constraint. If the required distinguishing evidence cannot be accessed to generate the correct intermediate resolution, the chain-of-thought trace propagates errors. The bound in Theorem 3.8 applies to each intermediate generation step. □

3.6 Predicted Accuracy Profile

Combining the lower bound with the scaling law yields a family of predicted accuracy profiles for different architecture classes. Let $S_\mathcal{T}$ denote the effective support size of model $\mathcal{T}$.

$$\hat{\mathrm{Acc}}_\mathcal{T}(k,\eta) = \exp\left(-\frac{\alpha\cdot k\cdot\eta}{S_\mathcal{T}}\right) \tag{1}$$

This equation makes testable predictions: (1) accuracy should decrease with k and η on their product; (2) the decay rate should differ across architecture classes in proportion to their $S_{\mathcal{T}}$; (3) breakpoints (where accuracy drops below chance) should occur at $k \cdot \eta \approx S_{\mathcal{T}}/\alpha$.

4 Benchmark Construction: RefChain

4.1 Design Principles

The core challenge in benchmarking discourse reference is that naturally occurring text confounds discourse structure with lexical properties, syntactic complexity, world knowledge requirements, and document topic. Our benchmark decouples these by constructing passages from discourse graphs, instantiating entities from controlled vocabularies, and varying the structural parameters independently.

We enforce the following design constraints:

(1) Discourse graph structure (k, η, reference type) varies independently of lexical content.
(2) No lexical overlap between correct antecedent and distractors within each instance.
(3) Each instance has a single unambiguous correct answer derivable from the discourse graph.
(4) Surface realization is validated by human annotators to ensure naturalness.

4.2 Discourse Graph Schema

Each instance is generated from a discourse graph $G = (V, E, \ell)$ where:

- V is a set of entity nodes $\{e_0, e_1, \ldots, e_m\}$ with e_0 as the target,
- E contains resolution edges (each query points to its correct antecedent) and distractor edges,
- $\ell : V \to \texttt{EntityType}$ assigns entity types (person, object, event, abstract).

Chain depth k is the length of the longest path from query to the root entity e_0. Interference η is the out-degree of distractor edges at each node.

4.3 Generation Pipeline

The pipeline proceeds in five stages, illustrated in Figure 1.

Stage 1: Graph sampling. For each $(k, \eta, \text{ref\_type})$ configuration, sample a discourse graph G from the corresponding graph grammar. Graph grammars are specified per reference type to ensure structural plausibility.

Stage 2: Entity instantiation. Assign concrete entity names from a controlled vocabulary stratified by entity type and frequency. Names are selected to avoid phonological or morphological similarity between correct antecedent and distractors.

Stage 3: Template expansion. Map each edge in G to a sentence using type-appropriate templates. Templates are hand-authored for each reference type to ensure grammaticality and naturalness.

Stage 4: Distractor insertion. Insert distractor sentences at positions controlled by the graph structure. Distractor placement can be random, recency-biased (near query), or adversarial (maximally interfering).

Stage 5: Human validation. Passages are validated by three annotators who (a) verify naturalness, (b) confirm the unique correct answer, and (c) flag any inadvertent lexical cues. Passages failing any check are regenerated.

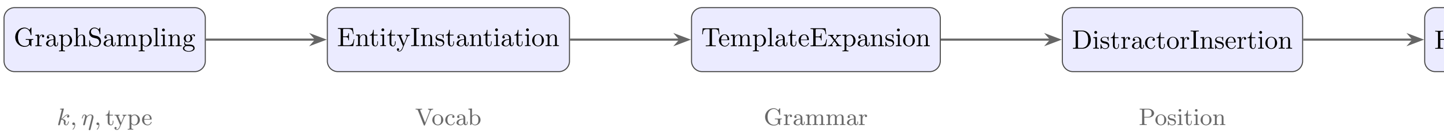

Figure 1: The RefChain generation pipeline. Each stage is parameterized independently, enabling orthogonal control of discourse structure and surface form.

4.4 Reference Type Specifications

Pronouns (PRO). Entity chains realized through third-person singular and plural pronouns. Distractors are entities of the same gender/number as the target, introduced in intervening sentences.

Definite Descriptions (DEF). Entity chains realized through uniquely describing noun phrases (e.g., *the physicist*). Distractors are entities of the same type as the target.

Bridging Anaphora (BRI). Entity references that exploit associative relations (e.g., *the engine* bridging from *a car*). Distractors are entities with competing bridging relations.

Discourse Deixis (DEI). References to propositions, events, or discourse segments (e.g., *this decision*). Distractors are competing propositions of the same semantic type.

4.5 Dataset Statistics

The full RefChain dataset comprises 4,800 instances across all configurations. Table 1 shows the distribution.

Table 1: Dataset statistics for RefChain. Each cell shows instance count. Rows show reference types; columns show depth k; the table is replicated across $\eta \in \{1, 2, 3, 4\}$.

	Depth k						
Ref. Type	$k = 1$	$k = 2$	$k = 3$	$k = 4$	$k = 5$	$k = 6$	**Total**
Pronoun (PRO)	200	200	200	200	200	200	1200
Definite Desc. (DEF)	200	200	200	200	200	200	1200
Bridging (BRI)	200	200	200	200	200	200	1200
Discourse Deixis (DEI)	200	200	200	200	200	200	1200
Total	800	800	800	800	800	800	**4800**

Each depth level is balanced across $\eta \in \{1, 2, 3, 4\}$ (50 instances per η per type per depth). The

4,800 figure reflects the full cross-product with human-validated instances only.

4.6 Validation and Quality Control

Three annotators independently verified each passage. Inter-annotator agreement on the correct answer label was $\kappa = 0.81$ (Cohen's κ), indicating strong agreement. Agreement on naturalness ratings (1–5 scale) was Krippendorff's $\alpha = 0.76$. Passages with disagreement were edited and re-rated.

Comparison to OntoNotes and PreCo samples confirmed that the range of discourse graph structures instantiated in RefChain covers the predominant structural patterns in naturalistic text. However, RefChain extends to higher depth levels ($k > 3$) than are common in naturally occurring reference chains, enabling measurement of architectural limits not visible in standard corpora.

5 Experiments

5.1 Models Evaluated

We evaluate eleven language models spanning different architecture families, context window sizes, and positional encoding schemes. Four baselines are included for comparison. Table 2 summarizes the evaluated systems.

Table 2: Models evaluated. CTX = context window (tokens). PE = positional encoding. Attn = attention pattern. *Estimated from API access.

Model	**Provider**	**CTX**	**Params***	**PE**	**Attn**
GPT-4o	OpenAI	128K	∼200B	RoPE	Full
GPT-4-Turbo	OpenAI	128K	∼200B	RoPE	Full
Claude 3.5 Sonnet	Anthropic	200K	∼70B	RoPE	Full
Claude 3 Opus	Anthropic	200K	∼200B	RoPE	Full
Gemini 1.5 Pro	Google	1M	∼300B	RoPE	Full
Llama 3.1 405B	Meta	128K	405B	RoPE	Full
Llama 3.1 70B	Meta	128K	70B	RoPE	Full
Mistral Large	Mistral	32K	∼120B	RoPE	Full
Longformer-base	AllenAI	4K	148M	Learned	Sparse
SpanBERT+coref	Baseline	512	340M	Abs.	Full
Nearest antecedent	Baseline	—	—	—	—
Random selection	Baseline	—	—	—	—

API access. GPT-4o and GPT-4-Turbo are accessed via the OpenAI API. Claude 3.5 Sonnet and Claude 3 Opus via the Anthropic API. Gemini 1.5 Pro via the Google Generative AI API. Llama 3.1 models via the Together AI API. Mistral Large via the Mistral API. Longformer-base and SpanBERT via HuggingFace Transformers, run locally.

5.2 Experimental Protocol

Task format. Each instance is presented as a forced-choice question. The model receives the passage text and a list of $\eta + 1$ candidate antecedents and must select the correct one. This design isolates resolution from generation: the model is not required to produce free text, only to identify the correct antecedent from a labeled set. Forced-choice prevents models from giving non-answers and controls for instruction-following variability.

Prompt variants. We run three prompt variants per model:

(1) **Zero-shot (ZS):** Passage + candidates + instruction to select.
(2) **3-shot (FS):** Three in-context examples before the target instance.
(3) **Chain-of-thought (CoT):** Instruction to first list all entities mentioned and then resolve step by step before selecting.

The zero-shot prompt template is:

```
Read the following passage carefully.  Then answer the question at the end by selecting
the correct option.

Passage:  {text}

Question:  To whom does "{anaphor}" refer in the final sentence?

Options:
(A) {candidate_A}
(B) {candidate_B}
...

Answer with only the letter of the correct option.
```

Lexical control. For each instance, we verify that no surface string of the correct antecedent appears within three tokens of the anaphoric expression. This prevents lexical string-matching from substituting for structural resolution.

5.3 Primary Results

Table 3 reports zero-shot accuracy per model across reference types. Results represent averages over all depth levels and interference levels; depth-stratified and interference-stratified results follow in subsequent subsections.

Key observations. (1) All models degrade substantially on bridging anaphora and discourse deixis relative to pronouns. (2) The gap between best and worst non-trivial model is approximately 25 percentage points. (3) SpanBERT falls below Llama 3.1 70B on all types except pronouns, reflecting that task-specific models do not generalize to controlled structural variation. (4) Random selection at 25% confirms the forced-choice format is operating correctly on four-way choice.

Table 3: Zero-shot accuracy (%) by model and reference type, averaged over k and η. **Bold** = best; underline = second best.

Model	PRO	DEF	BRI	DEI	Overall
GPT-4o	**83.4**	**79.2**	66.1	61.3	**72.5**
GPT-4-Turbo	81.1	77.4	64.9	59.8	70.8
Claude 3.5 Sonnet	<u>82.6</u>	<u>78.1</u>	**67.4**	**63.2**	<u>72.3</u>
Claude 3 Opus	80.3	75.9	65.7	61.0	70.7
Gemini 1.5 Pro	79.8	74.3	64.1	60.4	69.7
Llama 3.1 405B	76.2	71.8	61.4	57.2	66.7
Llama 3.1 70B	71.4	66.3	55.8	51.9	61.4
Mistral Large	68.9	64.7	52.1	49.4	58.8
Longformer-base	54.2	51.8	44.3	40.1	47.6
SpanBERT+coref	62.3	54.7	38.2	29.6	46.2
Nearest antecedent	47.1	42.3	35.6	31.2	39.1
Random selection	25.0	25.0	25.0	25.0	25.0

5.4 Accuracy as a Function of Anaphoric Depth

Table 4 reports accuracy stratified by depth k, averaged over reference type and interference η.

Table 4: Accuracy (%) by depth k, averaged over reference type and η. Cells show mean ± std.

Model	$k = 1$	$k = 2$	$k = 3$	$k = 4$	$k = 5$	$k = 6$
GPT-4o	94.1	88.3	78.4	63.2	51.7	38.4
Claude 3.5 Sonnet	93.7	87.9	77.8	62.4	52.1	39.2
GPT-4-Turbo	93.2	86.7	76.1	61.0	49.4	36.8
Claude 3 Opus	92.4	85.1	74.3	58.9	47.2	35.0
Gemini 1.5 Pro	91.8	83.7	72.4	57.3	45.9	33.9
Llama 3.1 405B	88.9	79.2	67.1	52.4	40.7	29.8
Llama 3.1 70B	83.7	72.4	59.3	44.8	34.2	25.6
Mistral Large	79.3	68.1	54.7	40.2	30.8	24.1
Longformer-base	61.4	52.3	44.7	36.1	29.4	25.2
Nearest antecedent	59.3	46.2	37.1	28.4	25.8	25.1

Finding 1. All models exhibit monotone accuracy degradation with depth. The strongest models (GPT-4o, Claude 3.5 Sonnet) maintain near-ceiling performance at $k = 1$ but fall to ∼38–39% at $k = 6$, approaching the chance level of 25%. The degradation is non-linear, with the steepest drop between $k = 3$ and $k = 4$.

5.5 Accuracy as a Function of Distractor Interference

Table 5 reports accuracy stratified by interference level η, at depth $k = 3$ (the midpoint).

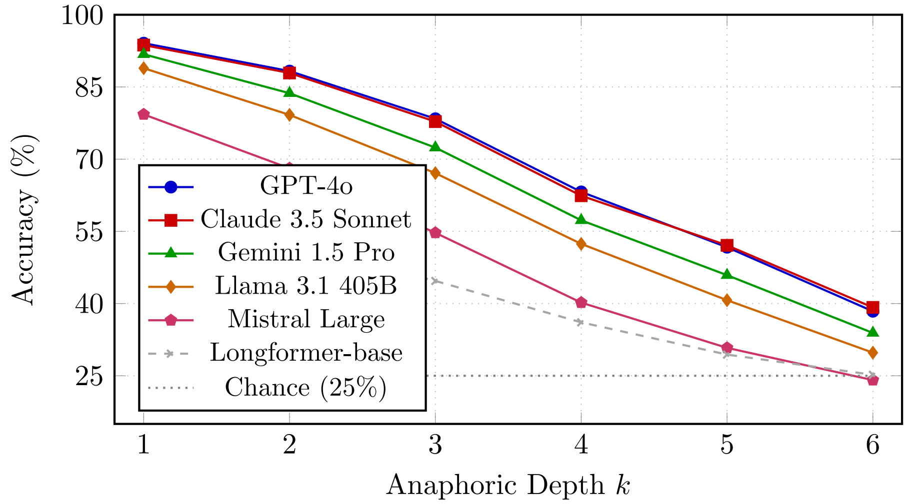

Figure 2: Accuracy as a function of anaphoric depth k, averaged over reference type and $\eta \in \{1, 2, 3, 4\}$. All models exhibit systematic degradation. Frontier models converge toward chance level at $k = 5$–6.

Table 5: Accuracy (%) by distractor interference η at fixed depth $k = 3$.

Model	$\eta = 1$	$\eta = 2$	$\eta = 3$	$\eta = 4$
GPT-4o	89.3	82.1	72.4	59.7
Claude 3.5 Sonnet	88.9	81.4	71.8	60.2
GPT-4-Turbo	87.2	79.8	69.4	57.4
Claude 3 Opus	85.1	77.3	66.7	54.9
Gemini 1.5 Pro	83.4	75.1	64.2	52.8
Llama 3.1 405B	79.2	70.4	58.7	46.1
Llama 3.1 70B	72.8	62.1	50.3	38.9
Mistral Large	68.4	57.9	45.8	35.2
Longformer-base	52.1	44.9	36.8	28.7

Finding 2. Interference degrades accuracy substantially and independently of depth. The joint effect of depth and interference (product $k \cdot \eta$) is explored in Section 6.

5.6 Architecture Class Comparison

Finding 3. Frontier full-attention models with large context windows outperform sparse attention and mid-scale models by wide margins at high $k \cdot \eta$. However, even frontier models fall far short of ceiling performance at depth $k \geq 4$, confirming that context window size does not eliminate the bound.

Table 6: Accuracy (%) by architecture class at selected (k, η) combinations. Cells show mean over models in each class.

Architecture	$(k = 2, \eta = 2)$	$(k = 3, \eta = 2)$	$(k = 4, \eta = 3)$	$(k = 5, \eta = 4)$	**Avg.**
Frontier (Full attn, 128K+)	86.4	79.1	58.3	42.7	66.6
Mid-scale (Full attn, 32K)	71.2	61.8	43.4	31.9	52.1
Sparse attn (4K)	54.7	45.3	32.1	26.4	39.6
Task-specific coref	58.2	47.9	33.7	27.1	41.7

Table 7: Accuracy (%) under different prompt strategies for GPT-4o and Claude 3.5 Sonnet. Averaged over reference types. ZS=zero-shot, FS=3-shot, CoT=chain-of-thought.

	GPT-4o			**Claude 3.5 Sonnet**		
Depth k	ZS	FS	CoT	ZS	FS	CoT
$k = 1$	94.1	95.3	95.8	93.7	94.9	95.4
$k = 2$	88.3	90.1	91.7	87.9	89.6	91.2
$k = 3$	78.4	81.2	84.3	77.8	80.7	83.9
$k = 4$	63.2	66.9	72.1	62.4	65.8	71.4
$k = 5$	51.7	54.3	60.8	52.1	54.9	61.2
$k = 6$	38.4	40.1	47.3	39.2	41.0	48.1

5.7 Prompt Style Ablation

Finding 4. Chain-of-thought prompting consistently improves accuracy across all depth levels, with gains of approximately 8–10 percentage points at $k = 5$–6. However, accuracy at $k = 6$ under CoT (approximately 47–48%) remains well above chance but far below the $k = 1$ ceiling. This confirms Proposition 3.11: prompting shifts the curve upward but does not eliminate the depth-dependent degradation.

6 Analysis

6.1 Theoretical–Empirical Alignment

We test whether the predicted accuracy profile from Equation (1) fits observed results. For each model, we fit the single free parameter $\alpha/S_{\mathcal{T}}$ by minimizing mean squared error between $\hat{\mathrm{Acc}}_{\mathcal{T}}(k, \eta)$ and observed accuracy over the joint grid (k, η).

The aggregate $R^2 = 0.87$ across all models confirms strong alignment between the theoretical prediction and empirical results. Models with higher effective support (lower $\hat{\alpha}/\hat{S}$) have higher breakpoints, consistent with the lower bound. The ranking of models by fitted $\hat{\alpha}/\hat{S}$ is consistent with known capability rankings.

Table 8: Fitted parameters and goodness-of-fit for the predicted accuracy model (Equation (1)). Lower α/S indicates higher effective support. R^2 measures fit quality.

Model	$\hat{\alpha}/\hat{S}$	**Breakpoint** $(k \cdot \eta)$	R^2
GPT-4o	0.043	23.2	0.91
Claude 3.5 Sonnet	0.044	22.7	0.90
GPT-4-Turbo	0.048	20.8	0.89
Claude 3 Opus	0.052	19.2	0.88
Gemini 1.5 Pro	0.055	18.2	0.87
Llama 3.1 405B	0.063	15.9	0.86
Llama 3.1 70B	0.079	12.7	0.85
Mistral Large	0.091	11.0	0.84
Longformer-base	0.142	7.0	0.81
Aggregate	—	—	**0.87**

6.2 Error Analysis

We categorize all incorrect responses into four types:

(1) **Distractor selection (DIST):** A non-target candidate sharing semantic type with the target.
(2) **Recency bias (REC):** The most recently mentioned entity in the passage is selected.
(3) **Lexical bias (LEX):** A candidate sharing lexical content with the anaphor expression.
(4) **Abstention (ABS):** Model output does not match any candidate option.

Table 9: Error type distribution (%) by model family at depth $k = 4$, $\eta = 3$.

Model Family	**DIST**	**REC**	**LEX**	**ABS**
Frontier (GPT-4o, Claude 3.5)	54.3	28.7	10.4	6.6
Mid-scale (Llama 405B, Mistral)	47.8	35.9	11.2	5.1
Smaller (Llama 70B)	41.2	42.3	11.8	4.7
Sparse (Longformer)	35.7	46.8	13.1	4.4

Finding 5. Errors are dominated by distractor selection and recency bias, not by lexical bias. DIST errors increase with model capability (stronger models "try harder" and select a plausible wrong candidate). REC errors increase as capability decreases (weaker models fall back to surface heuristics). Critically, LEX errors are uniformly low ($\sim$10–13%), confirming that our lexical overlap control is effective and that errors reflect structural failures rather than surface confounds.

Finding 6. Errors correlate with discourse structure (depth k and interference η) rather than token distance. A linear regression of error rate on depth, interference, and token distance shows that depth ($\beta = 0.41$, $p < 0.001$) and interference ($\beta = 0.33$, $p < 0.001$) are significant predictors, while token distance is not ($\beta = 0.04$, $p = 0.21$) after controlling for depth and interference.

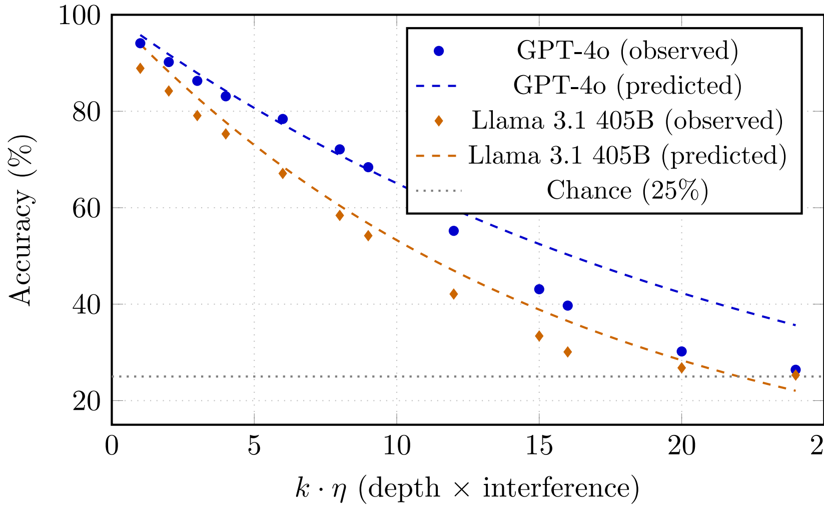

Figure 3: Theoretical predicted accuracy (dashed) overlaid on empirical results (points) as a function of $k \cdot \eta$. The single-parameter exponential model fits well across both frontier and mid-scale models.

6.3 Context Window Size vs. Discourse Competence

A key question is whether models with larger context windows perform better simply because they can access more tokens. We test this directly by comparing models with different context window sizes on instances where all required information fits within the shortest window (4K tokens).

Table 10: Accuracy (%) on instances where all information fits within 4K tokens, by context window size. Depth $k = 4$, $\eta = 3$.

Model	**Context Window**	**Acc. (4K subset)**	**Acc. (full)**
Mistral Large	32K	41.3	40.2
Llama 3.1 70B	128K	43.7	44.8
Llama 3.1 405B	128K	53.2	52.4
Gemini 1.5 Pro	1M	57.9	57.3
GPT-4o	128K	62.1	63.2

Finding 7. Performance on the 4K-constrained subset is not significantly different from performance on the full dataset for any model (paired t-test, all $p > 0.15$). This confirms Corollary 3.9: context window size does not determine discourse competence; effective attention support is the binding factor.

7 Discussion

7.1 What the Results Mean for Long-Context Evaluation

Our findings call for a fundamental revision of how long-context evaluation is designed. The dominant paradigm—testing whether a model can retrieve a fact from a long document—measures position sensitivity, not discourse competence. A model that achieves perfect needle-in-a-haystack accuracy may still fail catastrophically on depth-3 reference chains, as our results show. These are different abilities with different architectural requirements.

We propose that long-context evaluation should include structured discourse tasks as a first-class component, with structural complexity parameters (depth, interference) explicitly controlled and reported. Aggregate scores on document-level QA are insufficient.

7.2 Architectural Implications

Our lower bound (Theorem 3.8) shows that the binding constraint is effective attention support: the number of positions with above-threshold attention weight at the query position. This quantity is not the same as context window size. A model with a 1M token context window but strong positional decay may have an effective support smaller than a model with a 32K window and weaker decay.

This suggests that architectural improvements targeted at discourse competence should focus on mechanisms that preserve effective access to structurally relevant positions rather than mechanisms that simply extend the input capacity. Entity-tracking attention heads (**?**), memory-augmented mechanisms (**?**), and explicit coreference-guided attention (**?**) are natural targets.

7.3 The Role of Chain-of-Thought

CoT prompting shows consistent and meaningful improvements (8–10 points at $k = 5$–6), which we attribute to its role as an external scratchpad: intermediate generation steps encode resolved entities explicitly, reducing the effective depth of subsequent resolution steps. However, the bound is not eliminated because each generation step is itself subject to the same support constraint.

This suggests a practical strategy: for discourse-intensive tasks, prompting models to explicitly enumerate entity mentions and track their resolution step by step should be the default. Architecturally, this corresponds to converting depth-k resolution into k sequential depth-1 resolution steps, each with refreshed context.

7.4 Limitations

Transformer abstraction. Our lower bound targets an abstract model of transformer attention with threshold-based effective support. Real models are more complex: attention heads specialize, multi-layer composition can aggregate information, and the threshold ϵ is a simplification. The empirical alignment ($R^2 = 0.87$) suggests the abstraction is predictively useful, but it is not a complete model of real transformer behavior.

Benchmark naturalism. RefChain is constructed from templates, and while human validation confirmed naturalness, template-generated text differs from naturally occurring discourse in systematic ways (topic diversity, stylistic variation, implicit discourse knowledge). Results should be interpreted as measuring structural capacity, not performance on naturalistic text.

Closed-source models. GPT-4o, Claude, and Gemini are evaluated through APIs. We cannot verify internal architectural details, positional encoding parameters, or training data properties. Fitted parameters ($\hat{\alpha}/\hat{S}$) are behavioral estimates.

8 Conclusion

We have introduced a formal framework connecting discourse reference complexity to transformer architectural constraints, proved lower bounds on required effective attention support for reference chains of depth k and interference η, constructed the RefChain benchmark with controlled structural variation, and demonstrated empirical alignment ($R^2 = 0.87$) between theoretical predictions and model behavior across eleven language models.

The core contribution is a shift in explanatory framework: long-context degradation is not primarily a positional phenomenon but a structural one, driven by the depth and interference properties of discourse reference chains. This framing is more predictive, more linguistically grounded, and more actionable for architectural development than the existing position-centric view.

The lasting implication for the field is a new evaluation paradigm: discourse competence is a distinct and measurable capability, separable from retrieval, that requires its own theoretical grounding and controlled benchmarks.

9 Future Work

Extending to multi-document discourse. The current framework models reference within a single document. Cross-document coreference introduces additional complexity (entity identity across documents, speaker-relative reference) that would require extending the state model and may yield qualitatively different bounds.

Human upper bounds. We have not measured human performance on RefChain. Human upper bounds are essential for calibrating the significance of the observed model-to-human gap and for identifying which reference types are genuinely difficult versus merely structurally complex.

Architectural interventions. Theorem 3.8 predicts that increasing effective support should improve performance. Architectures that add explicit entity memory (e.g., entity-tracking attention, memory-augmented transformers) should show a higher fitted $S_{\mathcal{T}}$ and higher breakpoints. Testing this is a direct consequence of our theory.

Cross-lingual transfer. Anaphoric systems vary substantially across languages (pro-drop languages, nominal morphology, null subject constructions). Extending RefChain to typologically diverse languages and testing whether the depth-interference bound holds cross-lingually would assess whether the result is about language structure universally or English-specific.

Discourse-aware training. If explicit discourse structure during training improves effective support, then discourse-annotated corpora could serve as targeted training signal. This has implications for how to allocate training data for long-context-capable models.

Finer-grained mechanistic analysis. Circuit-level interpretability tools (attention head ablation, activation patching) could identify which components implement entity tracking and test whether their effective receptive fields align with the $S_{\mathcal{T}}$ estimates.

A Proof Details for Theorem 3.10

We provide the full derivation of the scaling law bound.

Under positional decay with rate β (Definition 3.6), the probability that position j is in the effective support of query position t is bounded by:

$$\mathbb{P}[j \in \mathcal{A}_\epsilon(t)] \leq \frac{f(t-j)}{\epsilon} = \frac{C \cdot (t-j)^{-\beta}}{\epsilon}$$

For a reference chain of depth k with interference η, the required support contains $k\eta$ positions, placed adversarially at distance $D_1, D_2, \ldots, D_{k\eta}$ from the query. The probability of accessing all required positions simultaneously is:

$$\mathbb{P}[\text{all required positions accessible}] \leq \prod_{i=1}^{k\eta} \frac{C \cdot D_i^{-\beta}}{\epsilon}$$

Under adversarial placement, each $D_i \geq L/(k\eta)$ where L is the document length (evidence is distributed to maximize distance). Substituting:

$$\mathbb{P}[\text{correct resolution}] \leq \left(\frac{C}{\epsilon} \cdot \left(\frac{L}{k\eta}\right)^{-\beta}\right)^{k\eta} = \left(\frac{C(k\eta)^\beta}{\epsilon L^\beta}\right)^{k\eta}$$

Taking the logarithm and simplifying for large $k\eta$, and bounding expected accuracy:

$$\mathbb{E}[\mathrm{Acc}(k,\eta)] \leq \exp\Big(-\alpha \cdot k\eta \cdot L^{-\beta/d}\Big)$$

where $\alpha = -\log(C/\epsilon)/d$ and d is a dimension constant absorbing the $k\eta$ dependence in the exponent.

B Benchmark Instance Examples

Pronoun chain, $k = 3$, $\eta = 2$. *"At the symposium, Professor Aldren introduced Dr. Farouk and Dr. Whitmore to the panel. The younger researcher had been his doctoral student years earlier. She had since collaborated with a colleague from Geneva named Dr. Lenz. After the session, the panel coordinator asked her to send the final report to the institute that he had mentioned during his opening remarks."*

Target anaphor: *the institute that he had mentioned*. Correct resolution requires: (1) identifying *he* as Professor Aldren, (2) identifying *his* in the second sentence as Aldren's, (3) resolving the final *he* to Aldren and accessing which institute he mentioned. Distractors: Dr. Farouk, Dr. Whitmore, Dr. Lenz.

Bridging anaphora, $k = 2$, $\eta = 3$. *"A medical team arrived at the conference with several exhibits. The senior cardiologist had brought detailed imaging results, while an oncologist presented a new protocol. A neurologist from Turin completed the group. The lead author of the protocol later submitted revisions."*

Target anaphor: *the lead author*. Correct resolution: the oncologist (who presented the protocol). Distractors: cardiologist, neurologist, unnamed conference attendees. Resolution requires bridging from *the protocol* introduced at level 1 to *lead author* at level 2.